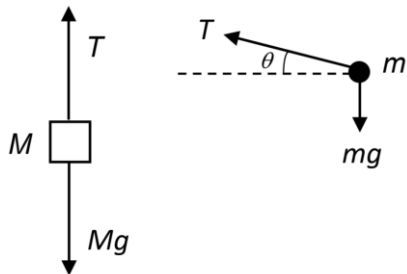


$$= 49\%$$

13



The horizontal component of tension  $T$  provides the centripetal force.

$$T = Mg \text{ .....(1)}$$

$$T \sin \theta = mg \text{ .....(2)}$$

$$T \cos \theta = m \frac{v^2}{r} \text{ .....(3) Solving using } \sin^2 \theta + \cos^2 \theta = 1, \quad M = m \sqrt{1 + \left( \frac{v^2}{rg} \right)^2}$$